Supplemental Material for 'Comment on "Canonical transcorrelated theory with projected Slater-type geminals" [J. Chem. Phys. 136, 084107 (2012)]'

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Here we document the reference expressions for the transcorrelated Hamiltonian described in 1. All notation and definitions used here exactly match Ref. 1, with one extension: primed indices $p'_1, p'_2, \ldots$ denote orbital basis set (OBS) that includes core orbitals even if the frozen-core (valence-only) approximation is invoked. As discussed in the main manuscript, the OBS range in Ref. 1 excludes core orbitals in the valence-only (frozen-core) computations. Both here and in Ref. 1 the definition of geminal operators involves valence occupied orbitals only (i.e., the geminal excitations do not occur from the core orbitals).

A complete expression for the transcorrelated Hamiltonian obtained *without the use of the cumulant decomposition* is:

$$\begin{aligned}
\hat{H} + [\hat{H}, \hat{A}^{\mathrm{F12}}] + \frac{1}{2}[[\hat{F}, \hat{A}^{\mathrm{F12}}], \hat{A}^{\mathrm{F12}}] = &\frac{1}{2} g^{p'_1 p'_2}_{\alpha_1\alpha_2} G^{\alpha_1\alpha_2}_{i_1 i_2} \hat{E}^{i_1 i_2}_{p'_1 p'_2} + h^{p'_1}_{x_1} G^{a_1 x_1}_{i_1 i_2} \hat{E}^{i_1 i_2}_{a_1 p'_1} \\
&+ \frac{1}{2} g^{\alpha_1\alpha_2}_{p'_1 p'_2} G^{i_1 i_2}_{\alpha_1\alpha_2} \hat{E}^{p'_1 p'_2}_{i_1 i_2} + h^{x_1}_{p'_1} G^{i_1 i_2}_{a_1 x_1} \hat{E}^{a_1 p'_1}_{i_1 i_2} \\
&+ g^{p'_1 x_1}_{p'_2 p'_3} G^{i_2 i_1}_{a_1 x_1} \hat{E}^{p'_3 a_1 p'_2}_{i_1 i_2 p'_1} + g^{p'_1 p'_3}_{p'_2 x_1} G^{x_1 a_1}_{i_1 i_2} \hat{E}^{i_2 p'_2 i_1}_{a_1 p'_1 p'_3} \\
&+ f^{x_2}_{x_1} G^{a_1 x_1}_{i_1 i_2} G^{i_4 i_3}_{a_2 x_2} \hat{E}^{i_2 a_2 i_1}_{i_3 i_4 a_1} - \frac{1}{2} f^{i_1}_{p'_1} G^{\alpha_1\alpha_2}_{i_1 i_2} G^{i_3 i_4}_{\alpha_1\alpha_2} \hat{E}^{p'_1 i_2}_{i_3 i_4} \\
&- \frac{1}{2} f^{p'_1}_{i_1} G^{\alpha_1\alpha_2}_{i_2 i_3} G^{i_4 i_1}_{\alpha_1\alpha_2} \hat{E}^{i_2 i_3}_{i_4 p'_1} + \frac{1}{2} f^{\alpha_1}_{p'_1} G^{a_1 x_1}_{i_1 i_2} G^{i_3 i_4}_{x_1\alpha_1} \hat{E}^{i_2 p'_1 i_1}_{i_3 i_4 a_1} \\
&+ \frac{1}{2} f^{p'_1}_{\alpha_1} G^{i_1 i_2}_{a_1 x_1} G^{x_1\alpha_1}_{i_3 i_4} \hat{E}^{a_1 i_3 i_4}_{i_1 i_2 p'_1} - \frac{1}{2} f^{p'_1}_{i_2} G^{i_1 i_2}_{a_1 x_1} G^{a_2 x_1}_{i_3 i_4} \hat{E}^{a_1 i_3 i_4}_{i_1 a_2 p'_1} \\
&- \frac{1}{2} f^{p'_1}_{i_2} G^{a_1 x_1}_{i_1 i_3} G^{i_2 i_4}_{a_2 x_1} \hat{E}^{i_3 i_1 a_2}_{i_4 a_1 p'_1} - \frac{1}{2} f^{i_2}_{p'_1} G^{a_1 x_1}_{i_1 i_2} G^{i_3 i_4}_{a_2 x_1} \hat{E}^{a_2 p'_1 i_1}_{i_3 i_4 a_1} \\
&- \frac{1}{2} f^{i_2}_{p'_1} G^{i_1 i_3}_{a_1 x_1} G^{a_2 x_1}_{i_2 i_4} \hat{E}^{a_1 i_4 p'_1}_{i_1 i_3 a_2} + f^{\alpha_2}_{\alpha_1} G^{\alpha_1\alpha_3}_{i_1 i_2} G^{i_4 i_3}_{\alpha_2\alpha_3} \hat{E}^{i_2 i_1}_{i_3 i_4}. \qquad \text{(S1)}
\end{aligned}$$

Note the appearance of the core-including OBS indices (p') in this expression. These expressions are identical to the spin-orbital expressions modulo the replacement of the spin-free normal-ordered (with respect to genuine vacuum) replacement operators $\hat{E}$ with the spin-orbital normal-ordered operators (a in the notation of Ref. 2), and can be used as to obtain cumulant-approximated transformed Hamiltonian expressions applicable to relativistic Hamiltonians and/or odd numbers of electrons.

Use of the spin-free cumulant-based approximation to the 3-body operator components produces the approximate transcorrelated Hamiltonian (Eq. (9) in Ref. 1):

$$\hat{\bar{H}}^{\mathrm{F12}} \equiv \hat{H} + [\hat{H}, \hat{A}^{\mathrm{F12}}]_{1,2} + \frac{1}{2}[[\hat{F}, \hat{A}^{\mathrm{F12}}]_{1,2}, \hat{A}^{\mathrm{F12}}]_{1,2} = \bar{h}^{p'}_{q'} \hat{E}^{q'}_{p'} + \frac{1}{2} \bar{g}^{p'r'}_{q's'} \hat{E}^{q's'}_{p'r'} \qquad \text{(S2)}$$

Expressions for the 1- and 2-body matrix elements of the approximate transcorrelated Hamil-

tonian (Eqs. (14-15) in Ref. 1, respectively) are:

$$
\begin{aligned}
\bar{h}^{p'}_{q'} =& h^{p'}_{q'} - \frac{1}{2} D^{p'_3}_{p'_2} D^{p'_1}_{i_1} G^{i_1 i_2}_{a_1 x_1} g^{p'_2 x_1}_{p'_1 p'_3} \delta^{a_1}_{q'} \delta^{p'}_{i_2} \\
& - 2 D^{p'_2}_{p'_3} D^{p'_1}_{i_1} G^{i_2 i_1}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{i_2} + D^{p'_2}_{p'_3} D^{i_1}_{p'_1} G^{x_1 a_1}_{i_1 i_2} g^{p'_1 p'_3}_{p'_2 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{2} D^{p'_2}_{i_2} D^{p'_1}_{i_1} G^{i_1 i_2}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{p'_3} - \frac{1}{2} D^{i_2}_{p'_2} D^{i_1}_{p'_1} G^{a_1 x_1}_{i_1 i_2} g^{p'_2 p'_1}_{p'_3 x_1} \delta^{p'_3}_{q'} \delta^{p'}_{a_1} \\
& + D^{i_2}_{p'_2} D^{i_1}_{p'_1} G^{x_1 a_1}_{i_1 i_2} g^{p'_2 p'_1}_{p'_3 x_1} \delta^{p'_3}_{q'} \delta^{p'}_{a_1} - \frac{1}{2} D^{p'_2}_{p'_3} D^{i_1}_{p'_1} G^{a_1 x_1}_{i_1 i_2} g^{p'_1 p'_3}_{p'_2 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} \\
& + D^{i_1}_{p'_3} D^{p'_1}_{p'_2} G^{a_1 x_1}_{i_1 i_2} g^{p'_2 p'_3}_{p'_1 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} + D^{p'_2}_{i_2} D^{p'_1}_{i_1} G^{i_2 i_1}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{p'_3} \\
& + D^{p'_2}_{p'_3} D^{p'_1}_{i_1} G^{i_1 i_2}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{i_2} - 2 D^{i_1}_{p'_3} D^{p'_1}_{p'_2} G^{x_1 a_1}_{i_1 i_2} g^{p'_2 p'_3}_{p'_1 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} \\
& + D^{p'_3}_{p'_2} D^{p'_1}_{i_1} G^{i_2 i_1}_{a_1 x_1} g^{p'_2 x_1}_{p'_1 p'_3} \delta^{a_1}_{q'} \delta^{p'}_{i_2} - \frac{1}{2} D^{p'_1 i_1}_{p'_2 p'_3} G^{a_1 x_1}_{i_1 i_2} g^{p'_2 p'_3}_{p'_1 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} \\
& + D^{p'_1 p'_2}_{i_1 p'_3} G^{i_2 i_1}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{i_2} - \frac{1}{2} D^{p'_1 p'_2}_{i_1 i_2} G^{i_2 i_1}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{p'_3} \\
& - \frac{1}{2} D^{p'_1 p'_2}_{i_1 p'_3} G^{i_1 i_2}_{a_1 x_1} g^{x_1 p'_3}_{p'_1 p'_2} \delta^{a_1}_{q'} \delta^{p'}_{i_2} - \frac{1}{2} D^{i_1 i_2}_{p'_1 p'_2} G^{x_1 a_1}_{i_1 i_2} g^{p'_2 p'_1}_{p'_3 x_1} \delta^{p'_3}_{q'} \delta^{p'}_{a_1} \\
& + D^{p'_1 i_1}_{p'_2 p'_3} G^{x_1 a_1}_{i_1 i_2} g^{p'_2 p'_3}_{p'_1 x_1} \delta^{i_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{4} G^{i_4 i_2}_{a_3 x_1} G^{a_1 x_1}_{i_1 i_3} f^{a_2}_{a_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_3}_{q'} \delta^{p'}_{a_2} - \frac{1}{2} G^{i_2 i_4}_{a_2 x_1} G^{x_1 a_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{2} G^{i_4 i_2}_{a_2 x_1} G^{a_1 x_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} - \frac{1}{4} G^{i_2 i_4}_{a_1 x_1} G^{x_1 a_3}_{i_1 i_3} f^{a_1}_{a_2} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_3} \\
& + \frac{1}{2} G^{i_2 i_4}_{a_3 x_1} G^{a_1 x_1}_{i_1 i_3} f^{a_2}_{a_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_3}_{q'} \delta^{p'}_{a_2} + \frac{1}{4} G^{i_2 i_5}_{a_2 x_1} G^{x_1 a_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{2} G^{i_2 i_5}_{a_2 x_1} G^{a_1 x_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} + \frac{1}{2} G^{i_2 i_4}_{a_1 x_1} G^{a_3 x_1}_{i_1 i_3} f^{a_1}_{a_2} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_3} \\
& + \frac{1}{4} G^{i_4 i_2}_{a_2 x_1} G^{x_1 a_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} - \frac{1}{2} G^{i_5 i_2}_{a_2 x_1} G^{x_1 a_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& + \frac{1}{4} G^{i_2 i_4}_{a_2 x_1} G^{a_1 x_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} + G^{i_4 i_2}_{a_2 x_2} G^{x_1 a_1}_{i_1 i_3} f^{x_2}_{x_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{2} G^{i_4 i_2}_{a_2 x_2} G^{a_1 x_1}_{i_1 i_3} f^{x_2}_{x_1} D^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} + \frac{1}{4} G^{i_5 i_2}_{a_2 x_1} G^{a_1 x_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& - \frac{1}{2} G^{i_4 i_2}_{a_2 x_2} G^{x_1 a_1}_{i_1 i_3} f^{x_2}_{x_1} D^{i_1 i_3}_{i_2 i_4} \delta^{a_2}_{q'} \delta^{p'}_{a_1} - \frac{1}{4} G^{i_2 i_4}_{a_3 x_1} G^{a_1 x_1}_{i_1 i_3} f^{a_2}_{a_1} D^{i_1 i_3}_{i_2 i_4} \delta^{a_3}_{q'} \delta^{p'}_{a_2} \\
& - \frac{1}{4} G^{i_2 i_4}_{a_1 x_1} G^{a_3 x_1}_{i_1 i_3} f^{a_1}_{a_2} D^{i_1 i_3}_{i_2 i_4} \delta^{a_2}_{q'} \delta^{p'}_{a_3} + \frac{1}{4} G^{i_5 i_2}_{a_2 x_1} G^{x_1 a_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_1 i_4}_{i_2 i_3} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& + \frac{1}{4} G^{i_2 i_4}_{a_2 x_1} G^{x_1 a_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_1 i_3}_{i_2 i_4} \delta^{a_2}_{q'} \delta^{p'}_{a_1} + \frac{1}{4} G^{i_4 i_2}_{a_2 x_1} G^{a_1 x_1}_{i_3 i_5} f^{i_5}_{i_1} D^{i_1 i_3}_{i_2 i_4} \delta^{a_2}_{q'} \delta^{p'}_{a_1} \\
& + \frac{1}{4} G^{i_2 i_5}_{a_2 x_1} G^{a_1 x_1}_{i_1 i_4} f^{i_3}_{i_5} D^{i_1 i_4}_{i_2 i_3} \delta^{a_2}_{q'} \delta^{p'}_{a_1}
\end{aligned}
\tag{S3}
$$

$$
\begin{aligned}
\bar{g}^{p'q'}_{r's'} =& \hat{S}(g^{p'q'}_{r's'} - D^{i_1}_{p'_1} G^{x_1a_1}_{i_1i_2} g^{p'_1p'_3}_{p'_2x_1} \delta^{i_2}_{r'} \delta^{p'_2}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_3} \\
&- D^{p'_1}_{i_1} G^{i_1i_2}_{a_1x_1} g^{p'_3x_1}_{p'_1p'_2} \delta^{p'_2}_{r'} \delta^{a_1}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{p'_3} - D^{p'_1}_{i_1} G^{i_2i_1}_{a_1x_1} g^{p'_3x_1}_{p'_1p'_2} \delta^{p'_2}_{r'} \delta^{a_1}_{s'} \delta^{p'}_{p'_3} \delta^{q'}_{i_2} \\
&+ 2D^{p'_2}_{p'_1} G^{x_1a_1}_{i_1i_2} g^{p'_1p'_3}_{p'_2x_1} \delta^{i_2}_{r'} \delta^{i_1}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_3} - D^{p'_1}_{p'_2} G^{x_1a_1}_{i_1i_2} g^{p'_3p'_2}_{p'_1x_1} \delta^{i_2}_{r'} \delta^{i_1}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_3} \\
&- D^{i_1}_{p'_1} G^{a_1x_1}_{i_1i_2} g^{p'_2p'_1}_{p'_3x_1} \delta^{p'_3}_{r'} \delta^{i_2}_{s'} \delta^{p'}_{p'_2} \delta^{q'}_{a_1} + 2D^{p'_1}_{p'_2} G^{i_1i_2}_{a_1x_1} g^{p'_2x_1}_{p'_1p'_3} \delta^{p'_3}_{r'} \delta^{a_1}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{i_1} \\
&+ 2D^{i_1}_{p'_1} G^{x_1a_1}_{i_1i_2} g^{p'_2p'_1}_{p'_3x_1} \delta^{i_2}_{r'} \delta^{p'_3}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_2} - D^{p'_1}_{i_1} G^{i_1i_2}_{a_1x_1} g^{x_1p'_2}_{p'_1p'_3} \delta^{a_1}_{r'} \delta^{p'_3}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{p'_2} \\
&- D^{p'_2}_{p'_1} G^{i_1i_2}_{a_1x_1} g^{x_1p'_1}_{p'_2p'_3} \delta^{a_1}_{r'} \delta^{p'_3}_{s'} \delta^{p'}_{i_1} \delta^{q'}_{i_2} + 2D^{p'_1}_{i_1} G^{i_2i_1}_{a_1x_1} g^{x_1p'_2}_{p'_1p'_3} \delta^{a_1}_{r'} \delta^{p'_3}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{p'_2} \\
&- D^{i_1}_{p'_1} G^{a_1x_1}_{i_1i_2} g^{p'_1p'_3}_{p'_2x_1} \delta^{p'_2}_{r'} \delta^{i_2}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_3} + 2G^{a_1x_1}_{i_1i_2} h^{p'_1}_{x_1} \delta^{i_1}_{r'} \delta^{i_2}_{s'} \delta^{p'}_{a_1} \delta^{q'}_{p'_1} \\
&+ V^{p'_1p'_2}_{i_1i_2} \delta^{i_1}_{r'} \delta^{i_2}_{s'} \delta^{p'}_{p'_1} \delta^{q'}_{p'_2} + \tilde{V}^{i_1i_2}_{p'_1p'_2} \delta^{p'_1}_{r'} \delta^{p'_2}_{s'} \delta^{p'}_{i_1} \delta^{q'}_{i_2} \\
&+ 2G^{i_1i_2}_{a_1x_1} h^{x_1}_{p'_1} \delta^{a_1}_{r'} \delta^{p'_1}_{s'} \delta^{p'}_{i_1} \delta^{q'}_{i_2} \\
&+ \frac{1}{2} G^{i_5i_2}_{a_2x_1} G^{a_1x_1}_{i_3i_4} f^{i_3}_{i_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} + \frac{1}{2} G^{i_5i_2}_{a_2x_1} G^{a_1x_1}_{i_1i_3} f^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&- G^{i_4i_2}_{a_2x_2} G^{a_1x_1}_{i_1i_3} f^{x_2}_{x_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} - G^{i_2i_4}_{a_2x_2} G^{x_1a_1}_{i_1i_3} f^{x_2}_{x_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} \\
&+ \frac{1}{2} G^{i_5i_3}_{a_2x_1} G^{a_1x_1}_{i_2i_4} f^{i_1}_{i_3} D^{i_2}_{i_1} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} - \frac{1}{2} G^{i_2i_4}_{a_3x_1} G^{a_1x_1}_{i_1i_3} f^{a_2}_{a_1} D^{i_1}_{i_2} \delta^{i_3}_{r'} \delta^{a_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_2} \\
&+ \frac{1}{2} G^{i_2i_3}_{a_2x_1} G^{a_1x_1}_{i_1i_5} f^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{i_5}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} - \frac{1}{2} G^{i_2i_4}_{a_1x_1} G^{a_3x_1}_{i_1i_3} f^{a_1}_{a_2} D^{i_1}_{i_2} \delta^{i_3}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_3} \\
&- \frac{1}{2} G^{i_2i_4}_{a_1x_1} G^{x_1a_3}_{i_1i_3} f^{a_1}_{a_2} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_3} - G^{i_5i_3}_{a_2x_1} G^{x_1a_1}_{i_2i_4} f^{i_1}_{i_3} D^{i_2}_{i_1} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&+ G^{i_4i_2}_{a_1x_1} G^{x_1a_3}_{i_1i_3} f^{a_1}_{a_2} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_3} - \frac{1}{2} G^{i_2i_4}_{a_3x_1} G^{x_1a_1}_{i_1i_3} f^{a_2}_{a_1} D^{i_1}_{i_2} \delta^{a_3}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_2} \\
&- \frac{1}{2} G^{i_4i_2}_{a_1x_1} G^{a_3x_1}_{i_1i_3} f^{a_1}_{a_2} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_3} + \frac{1}{2} G^{i_2i_5}_{a_2x_1} G^{a_1x_1}_{i_1i_3} f^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{i_4}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&- G^{i_5i_2}_{a_2x_1} G^{x_1a_1}_{i_3i_4} f^{i_3}_{i_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} - G^{i_3i_2}_{a_2x_1} G^{x_1a_1}_{i_1i_5} f^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_5}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} \\
&+ \frac{1}{2} G^{i_2i_5}_{a_2x_1} G^{x_1a_1}_{i_3i_4} f^{i_3}_{i_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} + \frac{1}{2} G^{i_3i_5}_{a_2x_1} G^{a_1x_1}_{i_2i_4} f^{i_1}_{i_3} D^{i_2}_{i_1} \delta^{i_4}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&- \frac{1}{2} G^{i_4i_2}_{a_3x_1} G^{a_1x_1}_{i_1i_3} f^{a_2}_{a_1} D^{i_1}_{i_2} \delta^{a_3}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_2} - G^{i_5i_2}_{a_2x_1} G^{x_1a_1}_{i_1i_3} f^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&+ G^{i_4i_2}_{a_3x_1} G^{x_1a_1}_{i_1i_3} f^{a_2}_{a_1} D^{i_1}_{i_2} \delta^{a_3}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_2} + \frac{1}{2} G^{i_2i_5}_{a_2x_1} G^{a_1x_1}_{i_3i_4} f^{i_3}_{i_1} D^{i_1}_{i_2} \delta^{i_4}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&+ \frac{1}{2} G^{i_2i_3}_{a_2x_1} G^{x_1a_1}_{i_1i_5} f^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_5}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} + \frac{1}{2} G^{i_3i_5}_{a_2x_1} G^{a_1x_1}_{i_2i_4} f^{i_1}_{i_3} D^{i_2}_{i_1} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&+ \frac{1}{2} G^{i_3i_2}_{a_2x_1} G^{a_1x_1}_{i_1i_5} f^{i_4}_{i_3} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_5}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} + 2G^{i_4i_2}_{a_2x_2} G^{x_1a_1}_{i_1i_3} f^{x_2}_{x_1} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_3}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} \\
&- G^{i_2i_4}_{a_2x_2} G^{a_1x_1}_{i_1i_3} f^{x_2}_{x_1} D^{i_1}_{i_2} \delta^{i_3}_{r'} \delta^{a_2}_{s'} \delta^{p'}_{i_4} \delta^{q'}_{a_1} + \frac{1}{2} G^{i_2i_5}_{a_2x_1} G^{x_1a_1}_{i_1i_3} f^{i_3}_{i_4} D^{i_1}_{i_2} \delta^{a_2}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_5} \delta^{q'}_{a_1} \\
&- X^{i_3i_1}_{i_4i_5} f^{i_2}_{i_1} \delta^{i_5}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{i_3} - X^{i_5i_3}_{i_1i_4} f^{i_1}_{i_2} \delta^{i_4}_{r'} \delta^{i_2}_{s'} \delta^{p'}_{i_3} \delta^{q'}_{i_5} \\
&+ B^{i_2i_3}_{i_1i_4} \delta^{i_1}_{r'} \delta^{i_4}_{s'} \delta^{p'}_{i_2} \delta^{q'}_{i_3}), \qquad \text{(S4)}
\end{aligned}
$$

where $\hat{S}$ is the particle symmetrization operator:

$$\hat{S}o^{p'q'}_{r's'} \equiv \frac{1}{2}(o^{p'q'}_{r's'} + o^{q'p'}_{s'r'}) \tag{S5}$$

and where we distinguish the usual nonhermitian F12 intermediate V (Eq. (23) in Ref. 1),

$$V^{p'_1p'_2}_{i_1i_2} \equiv g^{p'_1p'_2}_{\alpha\beta} G^{\alpha\beta}_{i_1i_2}, \tag{S6}$$

from its adjoint

$$\tilde{V}^{i_1i_2}_{p'_1p'_2} \equiv g^{\alpha\beta}_{p'_1p'_2} G^{i_1i_2}_{\alpha\beta} = \left(V^{p'_1p'_2}_{i_1i_2}\right)^* \neq V^{i_1i_2}_{p'_1p'_2}. \tag{S7}$$

Eqs. (2) to (4) can be verified using the open-source `SeQuant` software (see https://github.com/ValeevGroup/SeQuant/blob/e8067c72c4d8e8d3b8ddbd9eea8775244aadd1fa/examples/uccf12/uccf12.cpp).

Note the appearance of the primed (core-including) OBS indices in Eqs. (2) to (4). This is due to the exclusion of the core orbitals *only* in the definition of the geminal operator; this is referred to as frozen-core approach (**a**) in the main text. Ref. 1 implemented approach (**a+b1**) in which core orbitals are eliminated from the normal-ordered replacement operators in Eq. (1) *before* invoking the cumulant decomposition. As a concrete example consider $\hat{E}^{i_1i_2p'_1}_{p'_3a_1p'_2}$ appearing in the 6th term on the right-hand side of Eq. (1). In approach (**b1**) the initial elimination of the core orbitals ("fzc") turns it into $\hat{E}^{i_1i_2p_1}_{p_3a_1p_2}$, whose subsequent

cumulant decomposition ("cm") yields

$$
\begin{aligned}
\hat{E}^{i_1 i_2 p_1'}_{p_3' a_1 p_2'} \overset{\mathbf{b1}=\text{fzc+cm}}{\approx} \Bigg(& -\frac{1}{2} D^{i_2}_{p_3} \hat{E}^{p_1 i_1}_{p_2 a_1} + \frac{1}{2} D^{i_1}_{a_1} D^{p_1 i_2}_{p_2 p_3} \\
& + D^{p_1 i_2}_{p_2 a_1} \hat{E}^{i_1}_{p_3} - \frac{1}{2} D^{i_1}_{p_2} \hat{E}^{p_1 i_2}_{p_3 a_1} + D^{i_1}_{p_3} \hat{E}^{p_1 i_2}_{p_2 a_1} \\
& - D^{p_1}_{p_2} D^{i_1 i_2}_{p_3 a_1} - \frac{1}{2} D^{i_1 p_1}_{p_3 a_1} \hat{E}^{i_2}_{p_2} - D^{i_1}_{p_3} D^{p_1 i_2}_{p_2 a_1} \\
& + \frac{1}{2} D^{i_1}_{p_2} D^{p_1 i_2}_{p_3 a_1} - \frac{1}{2} D^{i_2}_{p_2} \hat{E}^{i_1 p_1}_{p_3 a_1} - \frac{1}{2} D^{p_1 i_2}_{p_3 a_1} \hat{E}^{i_1}_{p_2} \\
& + \frac{1}{2} D^{i_2}_{p_2} D^{i_1 p_1}_{p_3 a_1} - \frac{1}{2} D^{p_1 i_1}_{p_2 a_1} \hat{E}^{i_2}_{p_3} - \frac{1}{2} D^{i_2 i_1}_{p_2 p_3} \hat{E}^{p_1}_{a_1} \\
& - \frac{1}{2} D^{i_1 i_2}_{p_2 a_1} \hat{E}^{p_1}_{p_3} - \frac{1}{2} D^{p_1}_{a_1} \hat{E}^{i_2 i_1}_{p_2 p_3} - \frac{1}{2} D^{p_1}_{p_3} \hat{E}^{i_1 i_2}_{p_2 a_1} \\
& + D^{i_2}_{a_1} \hat{E}^{p_1 i_1}_{p_2 p_3} + D^{i_1 i_2}_{p_3 a_1} \hat{E}^{p_1}_{p_2} - \frac{1}{2} D^{i_1}_{a_1} \hat{E}^{p_1 i_2}_{p_2 p_3} \\
& + \frac{1}{2} D^{i_2}_{p_3} D^{p_1 i_1}_{p_2 a_1} + D^{p_1}_{p_2} \hat{E}^{i_1 i_2}_{p_3 a_1} + \frac{1}{2} D^{p_1}_{a_1} D^{i_2 i_1}_{p_2 p_3} \\
& + \frac{1}{2} D^{p_1}_{p_3} D^{i_1 i_2}_{p_2 a_1} - D^{i_2}_{a_1} D^{p_1 i_1}_{p_2 p_3} + D^{p_1 i_1}_{p_2 p_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{p_1 i_2}_{p_2 p_3} \hat{E}^{i_1}_{a_1} + 4 D^{p_1}_{p_2} D^{i_1}_{p_3} D^{i_2}_{a_1} - 2 D^{p_1}_{p_2} D^{i_2}_{a_1} \hat{E}^{i_1}_{p_3} \\
& - \frac{1}{2} D^{i_2}_{p_2} D^{i_1}_{a_1} \hat{E}^{p_1}_{p_3} + D^{p_1}_{p_3} D^{i_2}_{a_1} \hat{E}^{i_1}_{p_2} - 2 D^{i_1}_{p_3} D^{i_2}_{a_1} \hat{E}^{p_1}_{p_2} \\
& + D^{p_1}_{p_2} D^{i_2}_{p_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_2}_{p_2} D^{p_1}_{p_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_1}_{p_2} D^{p_1}_{a_1} \hat{E}^{i_2}_{p_3} \\
& - 2 D^{i_2}_{p_2} D^{i_1}_{p_3} D^{p_1}_{a_1} - \frac{1}{2} D^{p_1}_{p_3} D^{i_1}_{a_1} \hat{E}^{i_2}_{p_2} + D^{i_2}_{p_3} D^{i_1}_{a_1} \hat{E}^{p_1}_{p_2} \\
& + D^{i_2}_{p_2} D^{i_1}_{p_3} \hat{E}^{p_1}_{a_1} + D^{p_1}_{p_2} D^{i_1}_{a_1} \hat{E}^{i_2}_{p_3} + D^{i_1}_{p_3} D^{p_1}_{a_1} \hat{E}^{i_2}_{p_2} \\
& - 2 D^{p_1}_{p_2} D^{i_1}_{p_3} \hat{E}^{i_2}_{a_1} - \frac{1}{2} D^{i_1}_{p_2} D^{i_2}_{p_3} \hat{E}^{p_1}_{a_1} + D^{i_1}_{p_2} D^{i_2}_{p_3} D^{p_1}_{a_1} \\
& - 2 D^{p_1}_{p_2} D^{i_2}_{p_3} D^{i_1}_{a_1} + D^{i_2}_{p_2} D^{p_1}_{p_3} D^{i_1}_{a_1} + D^{i_1}_{p_2} D^{p_1}_{p_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{i_2}_{p_3} D^{p_1}_{a_1} \hat{E}^{i_1}_{p_2} - 2 D^{i_1}_{p_2} D^{p_1}_{p_3} D^{i_2}_{a_1} + D^{i_1}_{p_2} D^{i_2}_{a_1} \hat{E}^{p_1}_{p_3} \\
& + D^{i_2}_{p_2} D^{p_1}_{a_1} \hat{E}^{i_1}_{p_3} \Bigg)
\end{aligned}
\tag{S8}
$$

The other alternative, (**a+b2**), is obtained by invoking the cumulant decomposition first:

$$
\begin{aligned}
\hat{E}^{i_1 i_2 p'_1}_{p'_3 a_1 p'_2} \overset{\text{cm}}{\approx} \Bigg(& -\frac{1}{2} D^{i_2}_{p'_3} \hat{E}^{p'_1 i_1}_{p'_2 a_1} + \frac{1}{2} D^{i_1}_{a_1} D^{p'_1 i_2}_{p'_2 p'_3} \\
& + D^{p'_1 i_2}_{p'_2 a_1} \hat{E}^{i_1}_{p'_3} - \frac{1}{2} D^{i_1}_{p'_2} \hat{E}^{p'_1 i_2}_{p'_3 a_1} + D^{i_1}_{p'_3} \hat{E}^{p'_1 i_2}_{p'_2 a_1} \\
& - D^{p'_1}_{p'_2} D^{i_1 i_2}_{p'_3 a_1} - \frac{1}{2} D^{i_1 p'_1}_{p'_3 a_1} \hat{E}^{i_2}_{p'_2} - D^{i_1}_{p'_3} D^{p'_1 i_2}_{p'_2 a_1} \\
& + \frac{1}{2} D^{i_1}_{p'_2} D^{p'_1 i_2}_{p'_3 a_1} - \frac{1}{2} D^{i_2}_{p'_2} \hat{E}^{i_1 p'_1}_{p'_3 a_1} - \frac{1}{2} D^{p'_1 i_2}_{p'_3 a_1} \hat{E}^{i_1}_{p'_2} \\
& + \frac{1}{2} D^{i_2}_{p'_2} D^{i_1 p'_1}_{p'_3 a_1} - \frac{1}{2} D^{p'_1 i_1}_{p'_2 a_1} \hat{E}^{i_2}_{p'_3} - \frac{1}{2} D^{i_2 i_1}_{p'_2 p'_3} \hat{E}^{p'_1}_{a_1} \\
& - \frac{1}{2} D^{i_1 i_2}_{p'_2 a_1} \hat{E}^{p'_1}_{p'_3} - \frac{1}{2} D^{p'_1}_{a_1} \hat{E}^{i_2 i_1}_{p'_2 p'_3} - \frac{1}{2} D^{p'_1}_{p'_3} \hat{E}^{i_1 i_2}_{p'_2 a_1} \\
& + D^{i_2}_{a_1} \hat{E}^{p'_1 i_1}_{p'_2 p'_3} + D^{i_1 i_2}_{p'_3 a_1} \hat{E}^{p'_1}_{p'_2} - \frac{1}{2} D^{i_1}_{a_1} \hat{E}^{p'_1 i_2}_{p'_2 p'_3} \\
& + \frac{1}{2} D^{i_2}_{p'_3} D^{p'_1 i_1}_{p'_2 a_1} + D^{p'_1}_{p'_2} \hat{E}^{i_1 i_2}_{p'_3 a_1} + \frac{1}{2} D^{p'_1}_{a_1} D^{i_2 i_1}_{p'_2 p'_3} \\
& + \frac{1}{2} D^{p'_1}_{p'_3} D^{i_1 i_2}_{p'_2 a_1} - D^{i_2}_{a_1} D^{p'_1 i_1}_{p'_2 p'_3} + D^{p'_1 i_1}_{p'_2 p'_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{p'_1 i_2}_{p'_2 p'_3} \hat{E}^{i_1}_{a_1} + 4 D^{p'_1}_{p'_2} D^{i_1}_{p'_3} D^{i_2}_{a_1} - 2 D^{p'_1}_{p'_2} D^{i_2}_{a_1} \hat{E}^{i_1}_{p'_3} \\
& - \frac{1}{2} D^{i_2}_{p'_2} D^{i_1}_{a_1} \hat{E}^{p'_1}_{p'_3} + D^{p'_1}_{p'_3} D^{i_2}_{a_1} \hat{E}^{i_1}_{p'_2} - 2 D^{i_1}_{p'_3} D^{i_2}_{a_1} \hat{E}^{p'_1}_{p'_2} \\
& + D^{p'_1}_{p'_2} D^{i_2}_{p'_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_2}_{p'_2} D^{p'_1}_{p'_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_1}_{p'_2} D^{p'_1}_{a_1} \hat{E}^{i_2}_{p'_3} \\
& - 2 D^{i_2}_{p'_2} D^{i_1}_{p'_3} D^{p'_1}_{a_1} - \frac{1}{2} D^{p'_1}_{p'_3} D^{i_1}_{a_1} \hat{E}^{i_2}_{p'_2} + D^{i_2}_{p'_3} D^{i_1}_{a_1} \hat{E}^{p'_1}_{p'_2} \\
& + D^{i_2}_{p'_2} D^{i_1}_{p'_3} \hat{E}^{p'_1}_{a_1} + D^{p'_1}_{p'_2} D^{i_1}_{a_1} \hat{E}^{i_2}_{p'_3} + D^{i_1}_{p'_3} D^{p'_1}_{a_1} \hat{E}^{i_2}_{p'_2} \\
& - 2 D^{p'_1}_{p'_2} D^{i_1}_{p'_3} \hat{E}^{i_2}_{a_1} - \frac{1}{2} D^{i_1}_{p'_2} D^{i_2}_{p'_3} \hat{E}^{p'_1}_{a_1} + D^{i_1}_{p'_2} D^{i_2}_{p'_3} D^{p'_1}_{a_1} \\
& - 2 D^{p'_1}_{p'_2} D^{i_2}_{p'_3} D^{i_1}_{a_1} + D^{i_2}_{p'_2} D^{p'_1}_{p'_3} D^{i_1}_{a_1} + D^{i_1}_{p'_2} D^{p'_1}_{p'_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{i_2}_{p'_3} D^{p'_1}_{a_1} \hat{E}^{i_1}_{p'_2} - 2 D^{i_1}_{p'_2} D^{p'_1}_{p'_3} D^{i_2}_{a_1} + D^{i_1}_{p'_2} D^{i_2}_{a_1} \hat{E}^{p'_1}_{p'_3} \\
& + D^{i_2}_{p'_2} D^{p'_1}_{a_1} \hat{E}^{i_1}_{p'_3} \Bigg)
\end{aligned}
\tag{S9}
$$

The subsequent exclusion of the core orbitals replaces core-including OBS orbital indices on

the $\hat{E}$ operators only, but not on the RDMs:

$$\begin{aligned}
\hat{E}^{i_1 i_2 p'_1}_{p'_3 a_1 p'_2} \overset{\approx}{\mathbf{b2}=\mathrm{cm+fzc}} \Bigg(& -\frac{1}{2} D^{i_2}_{p'_3} \hat{E}^{p_1 i_1}_{p_2 a_1} + \frac{1}{2} D^{i_1}_{a_1} D^{p'_1 i_2}_{p'_2 p'_3} \\
& + D^{p'_1 i_2}_{p'_2 a_1} \hat{E}^{i_1}_{p_3} - \frac{1}{2} D^{i_1}_{p'_2} \hat{E}^{p_1 i_2}_{p_3 a_1} + D^{i_1}_{p'_3} \hat{E}^{p_1 i_2}_{p_2 a_1} \\
& - D^{p'_1}_{p'_2} D^{i_1 i_2}_{p'_3 a_1} - \frac{1}{2} D^{i_1 p'_1}_{p'_3 a_1} \hat{E}^{i_2}_{p_2} - D^{i_1}_{p'_3} D^{p'_1 i_2}_{p'_2 a_1} \\
& + \frac{1}{2} D^{i_1}_{p'_2} D^{p'_1 i_2}_{p'_3 a_1} - \frac{1}{2} D^{i_2}_{p'_2} \hat{E}^{i_1 p_1}_{p_3 a_1} - \frac{1}{2} D^{p'_1 i_2}_{p'_3 a_1} \hat{E}^{i_1}_{p_2} \\
& + \frac{1}{2} D^{i_2}_{p'_2} D^{i_1 p'_1}_{p'_3 a_1} - \frac{1}{2} D^{p'_1 i_1}_{p'_2 a_1} \hat{E}^{i_2}_{p_3} - \frac{1}{2} D^{i_2 i_1}_{p'_2 p'_3} \hat{E}^{p_1}_{a_1} \\
& - \frac{1}{2} D^{i_1 i_2}_{p'_2 a_1} \hat{E}^{p_1}_{p_3} - \frac{1}{2} D^{p'_1}_{a_1} \hat{E}^{i_2 i_1}_{p_2 p_3} - \frac{1}{2} D^{p'_1}_{p'_3} \hat{E}^{i_1 i_2}_{p_2 a_1} \\
& + D^{i_2}_{a_1} \hat{E}^{p_1 i_1}_{p_2 p_3} + D^{i_1 i_2}_{p'_3 a_1} \hat{E}^{p_1}_{p_2} - \frac{1}{2} D^{i_1}_{a_1} \hat{E}^{p_1 i_2}_{p_2 p_3} \\
& + \frac{1}{2} D^{i_2}_{p'_3} D^{p'_1 i_1}_{p'_2 a_1} + D^{p'_1}_{p'_2} \hat{E}^{i_1 i_2}_{p_3 a_1} + \frac{1}{2} D^{p'_1}_{a_1} D^{i_2 i_1}_{p'_2 p'_3} \\
& + \frac{1}{2} D^{p'_1}_{p'_3} D^{i_1 i_2}_{p'_2 a_1} - D^{i_2}_{a_1} D^{p'_1 i_1}_{p'_2 p'_3} + D^{p'_1 i_1}_{p'_2 p'_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{p'_1 i_2}_{p'_2 p'_3} \hat{E}^{i_1}_{a_1} + 4 D^{p'_1}_{p'_2} D^{i_1}_{p'_3} D^{i_2}_{a_1} - 2 D^{p'_1}_{p'_2} D^{i_2}_{a_1} \hat{E}^{i_1}_{p_3} \\
& - \frac{1}{2} D^{i_2}_{p'_2} D^{i_1}_{a_1} \hat{E}^{p_1}_{p_3} + D^{p'_1}_{p'_3} D^{i_2}_{a_1} \hat{E}^{i_1}_{p_2} - 2 D^{i_1}_{p'_3} D^{i_2}_{a_1} \hat{E}^{p_1}_{p_2} \\
& + D^{p'_1}_{p'_2} D^{i_2}_{p'_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_2}_{p'_2} D^{p'_1}_{p'_3} \hat{E}^{i_1}_{a_1} - \frac{1}{2} D^{i_1}_{p'_2} D^{p'_1}_{a_1} \hat{E}^{i_2}_{p_3} \\
& - 2 D^{i_2}_{p'_2} D^{i_1}_{p'_3} D^{p'_1}_{a_1} - \frac{1}{2} D^{p'_1}_{p'_3} D^{i_1}_{a_1} \hat{E}^{i_2}_{p_2} + D^{i_2}_{p'_3} D^{i_1}_{a_1} \hat{E}^{p_1}_{p_2} \\
& + D^{i_2}_{p'_2} D^{i_1}_{p'_3} \hat{E}^{p_1}_{a_1} + D^{p'_1}_{p'_2} D^{i_1}_{a_1} \hat{E}^{i_2}_{p_3} + D^{i_1}_{p'_3} D^{p'_1}_{a_1} \hat{E}^{i_2}_{p_2} \\
& - 2 D^{p'_1}_{p'_2} D^{i_1}_{p'_3} \hat{E}^{i_2}_{a_1} - \frac{1}{2} D^{i_1}_{p'_2} D^{i_2}_{p'_3} \hat{E}^{p_1}_{a_1} + D^{i_1}_{p'_2} D^{i_2}_{p'_3} D^{p'_1}_{a_1} \\
& - 2 D^{p'_1}_{p'_2} D^{i_2}_{p'_3} D^{i_1}_{a_1} + D^{i_2}_{p'_2} D^{p'_1}_{p'_3} D^{i_1}_{a_1} + D^{i_1}_{p'_2} D^{p'_1}_{p'_3} \hat{E}^{i_2}_{a_1} \\
& - \frac{1}{2} D^{i_2}_{p'_3} D^{p'_1}_{a_1} \hat{E}^{i_1}_{p_2} - 2 D^{i_1}_{p'_2} D^{p'_1}_{p'_3} D^{i_2}_{a_1} + D^{i_1}_{p'_2} D^{i_2}_{a_1} \hat{E}^{p_1}_{p_3} \\
& + D^{i_2}_{p'_2} D^{p'_1}_{a_1} \hat{E}^{i_1}_{p_3} \Bigg)
\end{aligned} \tag{S10}$$

Note the appearance of the core-including OBS indices in the (**b2**) approximant for $\hat{E}^{i_1 i_2 p'_1}_{p'_3 a_1 p'_2}$ (Eq. (10)) but not in the in the corresponding (**b1**) expression (Eq. (8)).

References

[1] T. Yanai and T. Shiozaki, "Canonical transcorrelated theory with projected Slater-type geminals," J. Chem. Phys. **136**, 084107 (2012).

[2]W. Kutzelnigg, "Quantum chemistry in Fock space. I. The universal wave and energy operators," J. Chem. Phys. **77**, 3081–3097 (1982).